

Rust coverage scale and weekly log

Laminate the scale and keep it with the jars. Five minutes at the start of each lesson: photograph, rate, record. No discussion.

Rating the coverage

Score	Name	What you can see
0	None	Nail looks the same as the day it went in. Still shiny.
1	Traces	A few orange specks or a faint tinge. You have to look for it.
2	Patchy	Obvious orange patches, but plenty of clean metal still showing.
3	Mostly covered	More rust than clean metal. Only small shiny areas left.
4	Fully covered	No clean metal visible. Rust may be flaking off into the liquid.

Rate all three jars separately Do not average them in your head. If your three jars disagree, that is data — write all three down and think about why they differ.

Weekly log

Week	Date	Jar 1	Jar 2	Jar 3	Anything you notice
Week 1 (set up)					
Week 2					
Week 3					
Week 4					
Week 5 (final)					

Week 5 only — the weigh-in

Take the nails out with tongs. Rinse, blot, and dry them properly before weighing — water left on a nail reads as rust and will invent a result that is not there.

	Jar 1	Jar 2	Jar 3
Starting mass (g)			
Final mass (g)			
Difference (g)			

If the nails are heavier than when they went in — where did the extra mass come from?

